

Week 1 #2

Set operations

Section 1.2.1

(a) *Intersection* of two sets $A \cap B$

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

(b) *Union* of two sets $A \cup B$

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

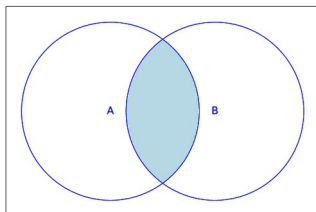
(c) A and B are *disjoint sets* $A \cap B = \emptyset$

(d) *The Universe, U* Superset of all sets.

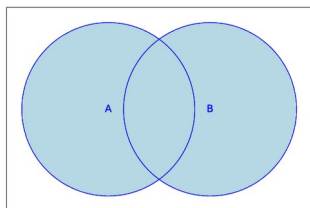
Section 1.2.2 Set Operations and their Venn Diagrams

When working with sets, as in other branches of mathematics, it is often quite useful to be able to draw a picture or diagram of the situation under consideration. A diagram of a set is called a Venn diagram. The universal set U is represented by the interior of a rectangle and the sets by disks inside the rectangle.

Venn diagram of the intersection and union of two sets.



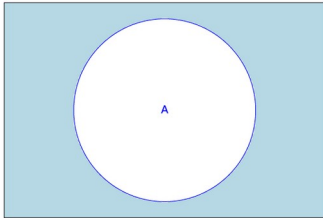
Intersection of two sets A and B



Union of two sets A and B

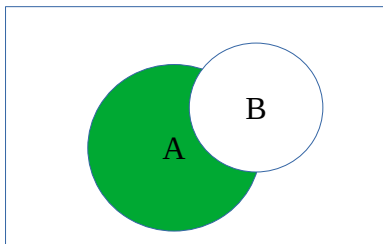
Complement of a set, A. Elements of the Universe not in A. Notation: A^c

$A^c = \{x \in U | x \notin A\}$. As a consequence of the definitions, $\emptyset^c = U$ and $U^c = \emptyset$



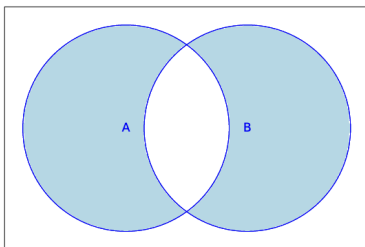
The subtraction operation $A - B$ between two sets A and B is defined as

$A - B = \{x \in A | x \notin B\}$. $A - B$ is the elements of A which are not in B. $B - A$ will be elements in B not in A.



The green represents $A - B$. This is elements that are in A but not in B

Let A and B be sets. The symmetric difference of A and B (denoted by $A \oplus B$) is the set of all elements that are in A and B but not in both. That is, $A \oplus B = (A \cup B) - (A \cap B)$



The blue area represents $A \oplus B$

Cartesian Product $A \times B$.

Let A and B be sets. *The Cartesian product of A and B*, denoted by $A \times B$, is defined as follows:

$$A \times B = \{(a, b) | a \in A \text{ and } b \in B\}$$

$A \times B$ is the set of all possible ordered pairs (a, b) whose first component comes from A and whose second component comes from B.

$$\text{If } A = \{2, 4, 7\} \text{ and } B = \{5, 9\}, \text{ then } A \times B = \{(2, 5), (2, 9), (4, 5), (4, 9), (7, 5), (7, 9)\}$$

If cardinality of a finite set A is m, and the cardinality of a finite set B is n, then $A \times B$ will have a cardinality of $m \times n$.

Extending this definition to 3 sets,

$A \times B \times C = \{(a, b, c) | a \in A \text{ and } b \in B \text{ and } c \in C\}$. This is the set of all triplets (a, b, c) such that a is from A, b is from B, and c is from C.

If A and B are the same sets, $A \times A = \{(a, b) | a \in A \text{ and } b \in A\}$. This is defined as A^2 for a set A.

Similarly $A \times A \times A = \{(a, b, c) | a \in A \text{ and } b \in A \text{ and } c \in A\} = A^3$, and for higher powers.

If set A has cardinality n, what is the cardinality of A^2 , of A^3 ?

Power set of a set

If A is any set, the set of all subsets of A is called *the power set of A*, denoted by $\mathcal{P}(A)$.

Example: If $A = \{a, b\}$, what are the subsets of A? By definition, the empty set is a subset of any set, and the set A itself is a subset of A. In addition, the sets $\{a\}$, $\{b\}$ are also subsets of A. So the set of all subsets of A, the power set of A, is $\mathcal{P}(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$

Example: If $A = \{4, 8, 6\}$, $\mathcal{P}(A) = \{\emptyset, \{4\}, \{8\}, \{6\}, \{4, 8\}, \{4, 6\}, \{8, 6\}, \{4, 8, 6\}\}$

1.3.4 Exercises

1. Let $A = \{0, 2, 3\}$, $B = \{2, 3\}$, $C = \{1, 4\}$, and let the universal set be $U = \{0, 1, 2, 3, 4\}$. List the elements of
 - (a) $A \times B$
 - (b) $B \times A$
 - (c) $A \times B \times C$
 - (d) $U \times \emptyset$
 - (e) $A \times A^c$
 - (f) B^2
 - (g) B^3
 - (h) $B \times \mathcal{P}(B)$
2. Suppose that you are about to flip a coin and then roll a die. Let $A = \{HEADS, TAILS\}$ and $B = \{1, 2, 3, 4, 5, 6\}$.
 - (a) What is $|A \times B|$?
 - (b) How could you interpret the set $A \times B$?
3. List all two-element sets in $\mathcal{P}(\{a, b, c, d\})$
4. List all three-element sets in $\mathcal{P}(\{a, b, c, d\})$.
5. How many singleton (one-element) sets are there in $\mathcal{P}(A)$ if $|A| = n$?
6. A person has four coins in his pocket: a penny, a nickel, a dime, and a quarter. How many different sums of money can he take out if he removes 3 coins at a time?
7. Let $A = \{+, -\}$ and $B = \{00, 01, 10, 11\}$.
 - (a) List the elements of $A \times B$
 - (b) How many elements do A^4 and $(A \times B)^3$ have?
8. Let $A = \{\bullet, \square, \otimes\}$ and $B = \{\square, \ominus, \bullet\}$.
 - (a) List the elements of $A \times B$ and $B \times A$. The parentheses and comma in an ordered pair are not necessary in cases such as this where the elements of each set are individual symbols.
 - (b) Identify the intersection of $A \times B$ and $B \times A$ for the case above, and then guess at a general rule for the intersection of $A \times B$ and $B \times A$, where A and B are any two sets.
9. Let A and B be nonempty sets. When are $A \times B$ and $B \times A$ equal?